# Steady convection and diffusion in one dimension

Step-by-step COMSOL Multiphysics 4.3a guide for students

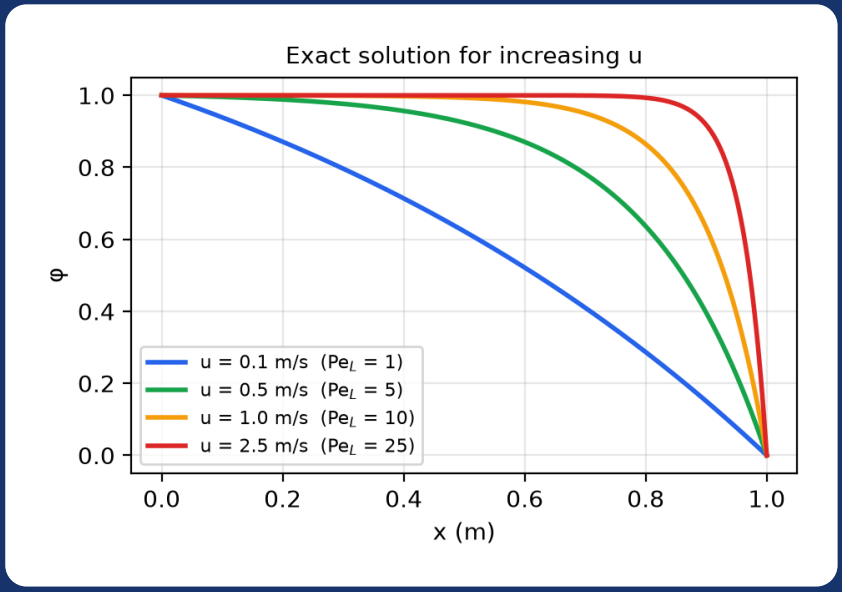
Problem: central versus upwind differences for the convection term (Introduction to CFD, Problem 5; Versteeg & Malalasekera, ch. 5)

Tool: COMSOL Multiphysics 4.3a, Transport of Diluted Species, 1D stationary study with stabilisation switched off

Result, case 2 (u = 2.5 m/s, Pe = 5 per cell): central differences give phi = 0.95, 1.06, 0.80, 1.41 at x = 0.2 ... 0.8; upwind 1.00, 1.00, 0.97, 0.83 (exact 1, 1, 1, 0.993)

COMSOL without stabilisation reproduces central differences to 1e-15; its default stabilisation removes the wiggles

Includes: hand solution (coefficients, Thomas algorithm, false diffusion, Gauss-Seidel), 16 annotated COMSOL steps, grid study



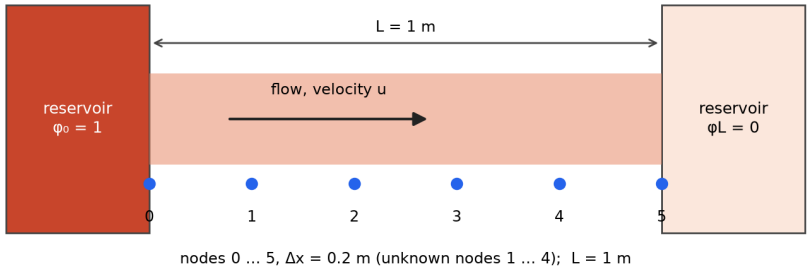

Model file: convdiff_comsol.mph (stabilisation off; open it in COMSOL to compare with your own model)

# The problem and the discretisation



A property phi (a temperature, or the concentration of a dye) is carried by convection and spreads by diffusion through a 1D domain 0 <= x <= L = 1 m between two reservoirs. The fluid flows from left to right with constant velocity u. The left reservoir holds phi = 1, the right one phi = 0. Find the steady distribution with the finite difference method, discretising the convection term with central and with upwind differences, for u = 0.1 m/s and u = 2.5 m/s. Grid: 5 intervals, Dx = 0.2 m. (Adapted from Versteeg & Malalasekera, chapter 5.)

| Quantity | Symbol | Value |
|---|---|---|
| length | L | 1 m |
| density | rho | 1 kg/m^3 |
| diffusion coefficient | Gamma | 0.1 kg/(m s) |
| left / right value | phi_0 / phi_L | 1 / 0 |
| velocity, case 1 / 2 | u | 0.1 / 2.5 m/s |
| grid, cases 1 and 2 | N, Dx | 5 intervals, 0.2 m |
| grids, case 3 | N | 5 ... 320, u = 2.5 |




nodes 0 ... 5, Δx = 0.2 m (unknown nodes 1 ... 4); L = 1 m

Assumptions: steady state; 1D; constant rho, u and Gamma; no source of phi.

## 1. Physics

$$\frac{d(\rho u \phi)}{dx} = \frac{d}{dx}\left(\Gamma \frac{d\phi}{dx}\right)$$

Convection (left) carries phi downstream with the flow; diffusion (right) spreads it in both directions, from high to low values. The ratio is the Peclet number:

$$Pe_L = \frac{\rho u L}{\Gamma} = 1 \text{ and } 25, \qquad Pe_\Delta = \frac{\rho u \Delta x}{\Gamma} = 0.2 \text{ and } 5$$

Exact solution: (phi - phi_0)/(phi_L - phi_0) = (exp(rho u x/Gamma) - 1)/(exp(rho u L/Gamma) - 1). For large u phi stays close to 1 over most of the domain and drops sharply to 0 in a thin layer at x = L.

## 2. Discretisation, F = rho u, D = Gamma / Dx

Integrate over a control volume (or use differences) and write a P phi_i = a W phi_(i-1) + a E phi_(i+1):

| scheme | a_W | a_E | a_P |
|---|---|---|---|
| central (CDS) | D + F/2 | D - F/2 | a_W + a_E = 2D |
| upwind (UDS), u > 0 | D + F | D | a_W + a_E = 2D + F |

CDS: dphi/dx = (phi_(i+1) - phi_(i-1))/(2 Dx). UDS: dphi/dx = (phi_i - phi_(i-1))/Dx for u > 0. For F = 2D (Pe = 2) the CDS coefficient a_E becomes zero; for larger Pe it is negative.

# Hand solution: coefficients and the Thomas algorithm

## A. Coefficients for the two velocities (Dx = 0.2 m: D = Gamma/Dx = 0.5)

| | u (m/s) | F = rho u | Pe_Delta = F/D | scheme | a_W | a_E | a_P | a_E sign |
|---|---|---|---|---|---|---|---|---|
| case 1 | 0.1 | 0.1 | 0.2 | CDS | 0.55 | 0.45 | 1.00 | positive |
| case 1 | 0.1 | 0.1 | 0.2 | UDS | 0.60 | 0.50 | 1.10 | positive |
| case 2 | 2.5 | 2.5 | 5.0 | CDS | 1.75 | -0.75 | 1.00 | NEGATIVE |
| case 2 | 2.5 | 2.5 | 5.0 | UDS | 3.00 | 0.50 | 3.50 | positive |

Interior equations (nodes 1 to 4, phi_0 = 1, phi_5 = 0): a_P phi_i - a_W phi_(i-1) - a_E phi_(i+1) = 0. This is a tridiagonal system: lower diagonal a_i = -a_W, main diagonal b_i = a_P, upper diagonal c_i = -a_E, right-hand side d_1 = a_W phi_0, d_2 = d_3 = 0, d_4 = a_E phi_5 = 0.

## B. Thomas algorithm, worked for case 2 (u = 2.5 m/s)

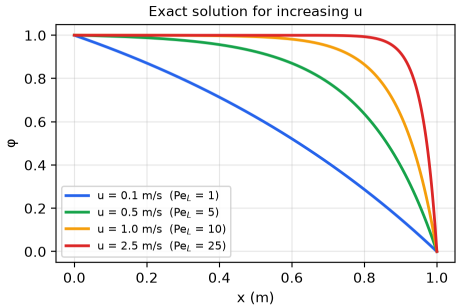
Forward sweep: c'_1 = c_1/b_1, d'_1 = d_1/b_1; c'_i = c_i/(b_i - a_i c'_(i-1)), d'_i = (d_i - a_i d'_(i-1))/(b_i - a_i c'_(i-1)). Back substitution: phi_4 = d'_4, phi_i = d'_i - c'_i phi_(i+1).

**CDS (a_W = 1.75, a_E = -0.75, a_P = 1.00):**

| i | a_i | b_i | c_i | d_i | c'_i | d'_i | phi_i |
|---|---|---|---|---|---|---|---|
| 1 | 0.000 | 1.000 | 0.750 | 1.750 | 0.7500 | 1.7500 | 0.9525 |
| 2 | -1.750 | 1.000 | 0.750 | 0.000 | 0.3243 | 1.3243 | 1.0633 |
| 3 | -1.750 | 1.000 | 0.750 | 0.000 | 0.4784 | 1.4784 | 0.8047 |
| 4 | -1.750 | 1.000 | 0.000 | 0.000 | 0.0000 | 1.4082 | 1.4082 |

**UDS (a_W = 3.00, a_E = 0.50, a_P = 3.50):**

| i | a_i | b_i | c_i | d_i | c'_i | d'_i | phi_i |
|---|---|---|---|---|---|---|---|
| 1 | 0.000 | 3.500 | -0.500 | 3.000 | -0.1429 | 0.8571 | 0.9994 |
| 2 | -3.000 | 3.500 | -0.500 | 0.000 | -0.1628 | 0.8372 | 0.9955 |
| 3 | -3.000 | 3.500 | -0.500 | 0.000 | -0.1660 | 0.8340 | 0.9723 |
| 4 | -3.000 | 3.500 | 0.000 | 0.000 | 0.0000 | 0.8334 | 0.8334 |




Exact solutions for several velocities: the faster the flow, the thinner the layer at x = 1.

# Hand solution: results, false diffusion and Gauss-Seidel

## A. Nodal values

| Case 1 (u = 0.1 m/s, Pe_Delta = 0.2) | x=0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | max error |
|---|---|---|---|---|---|---|---|
| exact | 1.0000 | 0.8711 | 0.7138 | 0.5215 | 0.2868 | 0.0000 | |
| CDS | 1.0000 | 0.8714 | 0.7141 | 0.5220 | 0.2871 | 0.0000 | 0.0004 |
| UDS | 1.0000 | 0.8656 | 0.7044 | 0.5109 | 0.2786 | 0.0000 | 0.0107 |

| Case 2 (u = 2.5 m/s, Pe_Delta = 5) | x=0 | 0.2 | 0.4 | 0.6 | 0.8 | 1.0 | max error |
|---|---|---|---|---|---|---|---|
| exact | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9933 | 0.0000 | |
| CDS | 1.0000 | 0.9525 | 1.0633 | 0.8047 | 1.4082 | 0.0000 | 0.4149 |
| UDS | 1.0000 | 0.9994 | 0.9955 | 0.9723 | 0.8334 | 0.0000 | 0.1598 |

## B. False diffusion (Task 6): the upwind scheme is the central scheme with a larger diffusion coefficient

UDS: a_W = D + F, a_E = D. CDS with Gamma' = Gamma + rho u Dx/2: a_W = D' + F/2 = D + F/2 + F/2 = D + F, a_E = D' - F/2 = D. The two sets are identical, so the UDS solution equals the CDS solution with the extra diffusion rho u Dx / 2 (we checked that the two solutions agree to 1e-16).

| case | rho u Dx / 2 | compared with Gamma = 0.1 |
|---|---|---|
| 1: u = 0.1 | 0.010 | 10 % of Gamma: small, UDS is accurate (max error 0.011) |
| 2: u = 2.5 | 0.250 | 250 % of Gamma: the numerical diffusion is 2.5 times the real one |

## C. Gauss-Seidel on the case 2 system (Task 8, optional): phi_i = (a_W phi_(i-1) + a_E phi_(i+1))/a_P, start from 0, order i = 1 ... 4

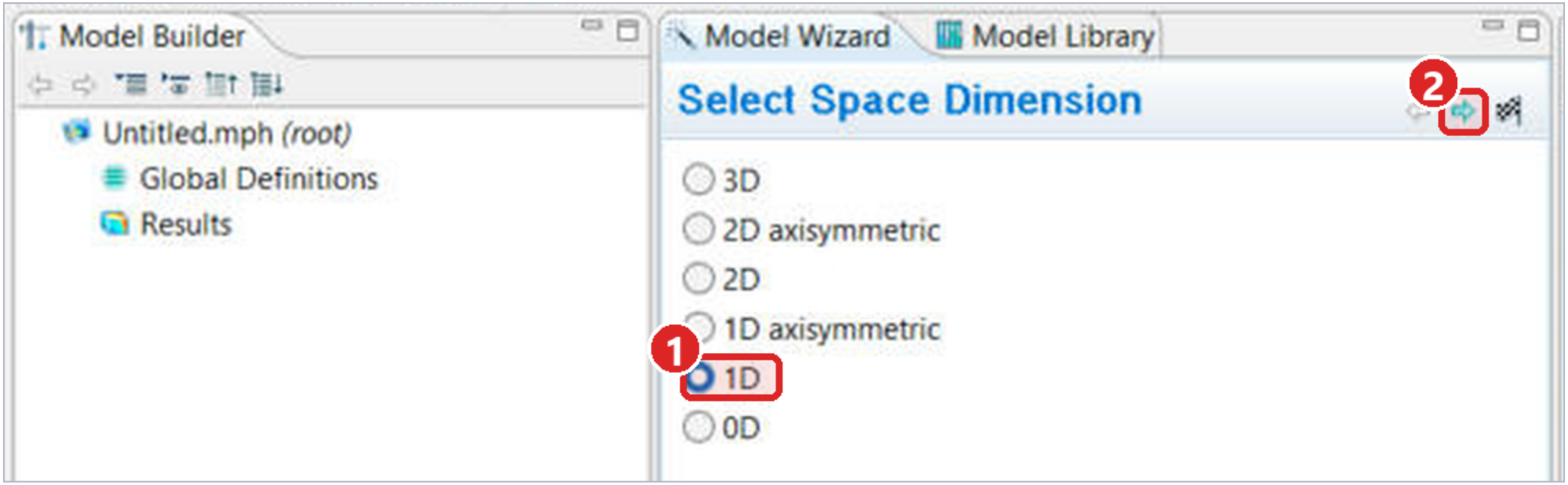
| iteration | 0 | 1 | 2 | 5 | 10 | 20 | 50 |
|---|---|---|---|---|---|---|---|
| max \|phi - direct\|, CDS | 1.4e+00 | 8.0e+00 | 2.9e+01 | 1.2e+03 | 5.7e+05 | 1.3e+11 | 1.6e+27 |
| max \|phi - direct\|, UDS | 1.0e+00 | 3.4e-01 | 1.1e-01 | 3.7e-03 | 1.2e-05 | 1.4e-10 | 1.1e-16 |

UDS converges quickly (spectral radius of the iteration matrix 0.32; the error falls by 1e-3 every 3 iterations). CDS diverges (spectral radius 3.44 > 1): the error grows by a factor of 3.4 per sweep. The reason is that a_E < 0 destroys the diagonal dominance (|a_W| + |a_E| = 2.5 > a_P = 1), which iterative methods need; the direct Thomas algorithm still solves the system, but the answer is wrong physically.

# Step 1 - Start a 1D model



*Start COMSOL Multiphysics 4.3a. The Model Wizard opens (if not: File > New).*



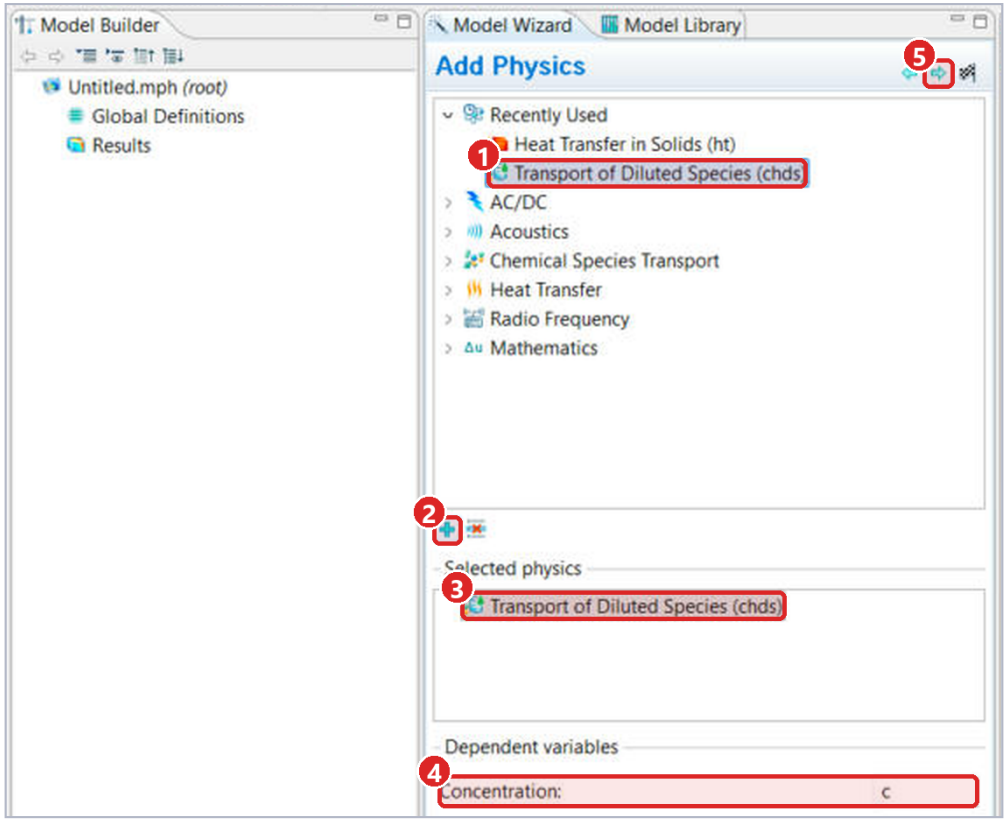

1. Select 1D. The problem is one-dimensional: phi changes only along x (Assumption 2).
2. Click the Next arrow.

**Why?**
A 1D model needs only a line: the left end is the reservoir with phi = 1, the right end the reservoir with phi = 0.

# Step 2 - Add Transport of Diluted Species



*Convection and diffusion of a property (a dye concentration, or a temperature) is described by the Transport of Diluted Species interface.*




1. Select Transport of Diluted Species (chds). It is in the folder Chemical Species Transport and also under Recently Used.
2. Click the blue + (Add Selected) button.
3. The interface appears under Selected physics.
4. The dependent variable is the concentration c (unit mol/m^3). The numbers are the same as for phi.
5. Click Next.

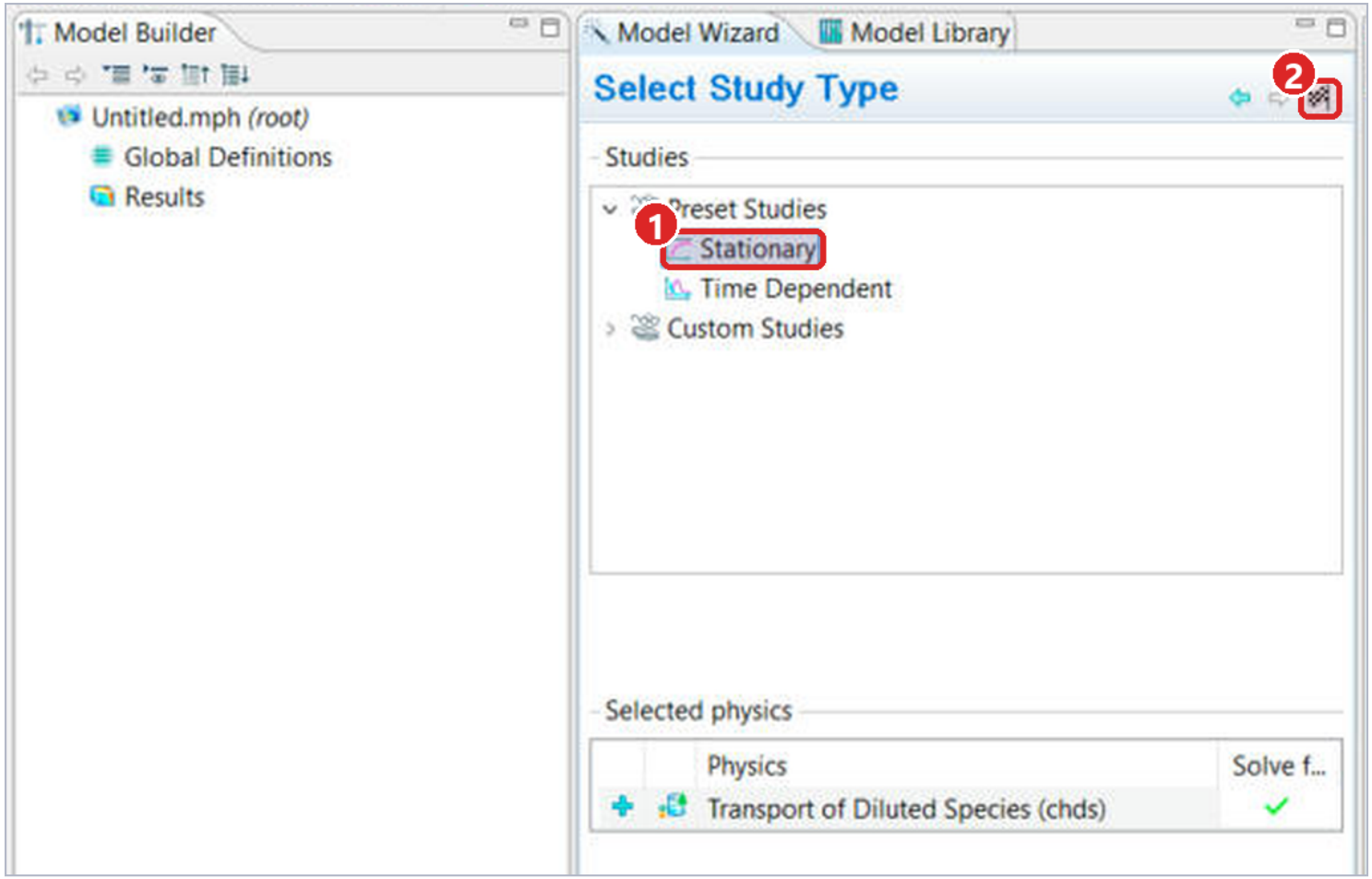
**Why?**

The interface solves div(-D grad c + u c) = R in steady state. With R = 0 and constant u in 1D this is u dc/dx = D d2c/dx2, the problem's equation divided by the density rho = 1 (D = Gamma / rho).

# Step 3 - Choose a stationary study



*Steady state means phi no longer changes with time (Assumption 1).*

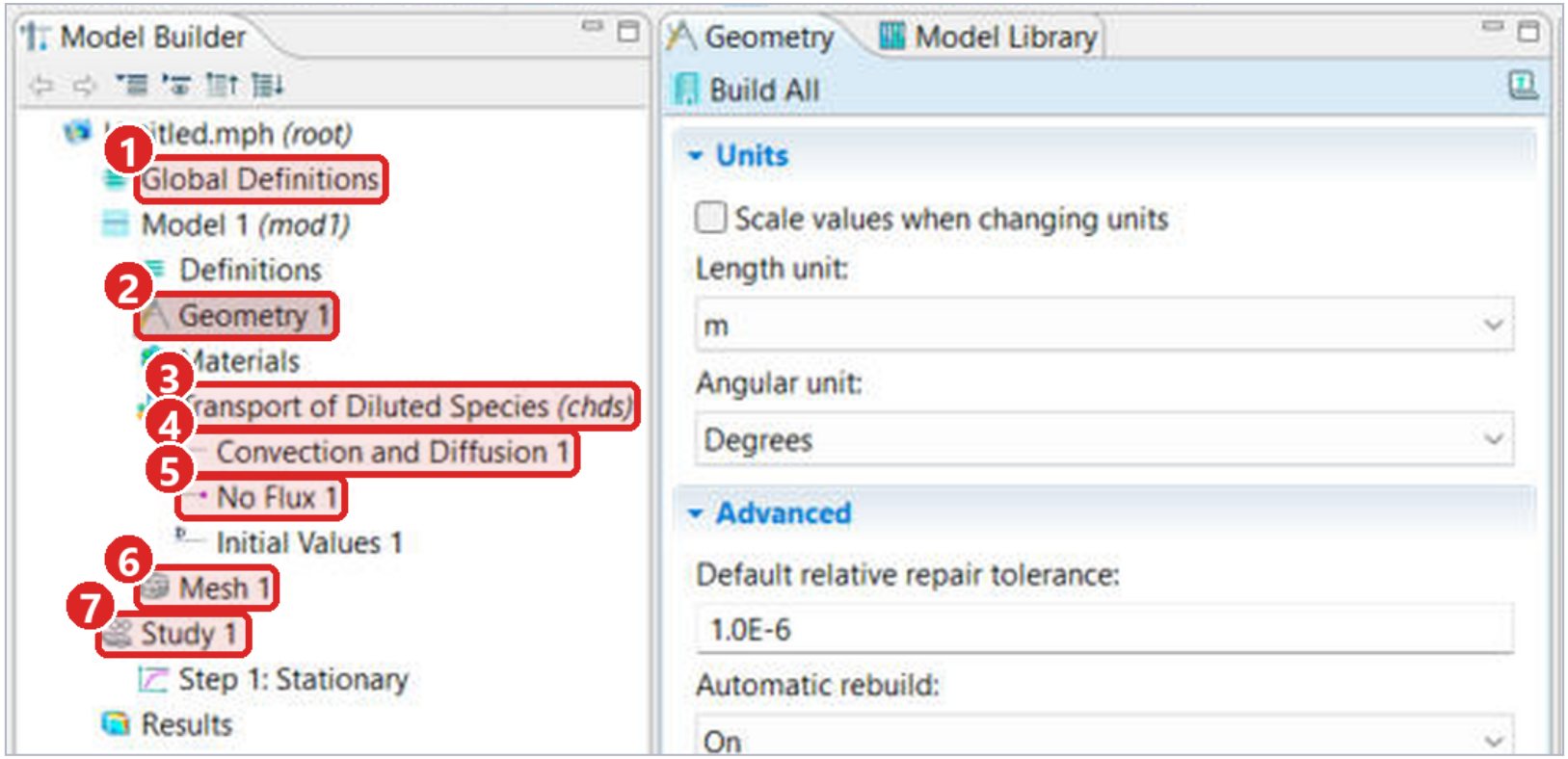



1. Select Stationary (under Preset Studies).
2. Click Finish (the chequered-flag icon).

**Why?**
A stationary study solves the whole system of node equations at once: the tridiagonal system that the Thomas algorithm solves by hand.

# Step 4 - Tour of the model tree

*After Finish the Model Builder holds the model outline.*



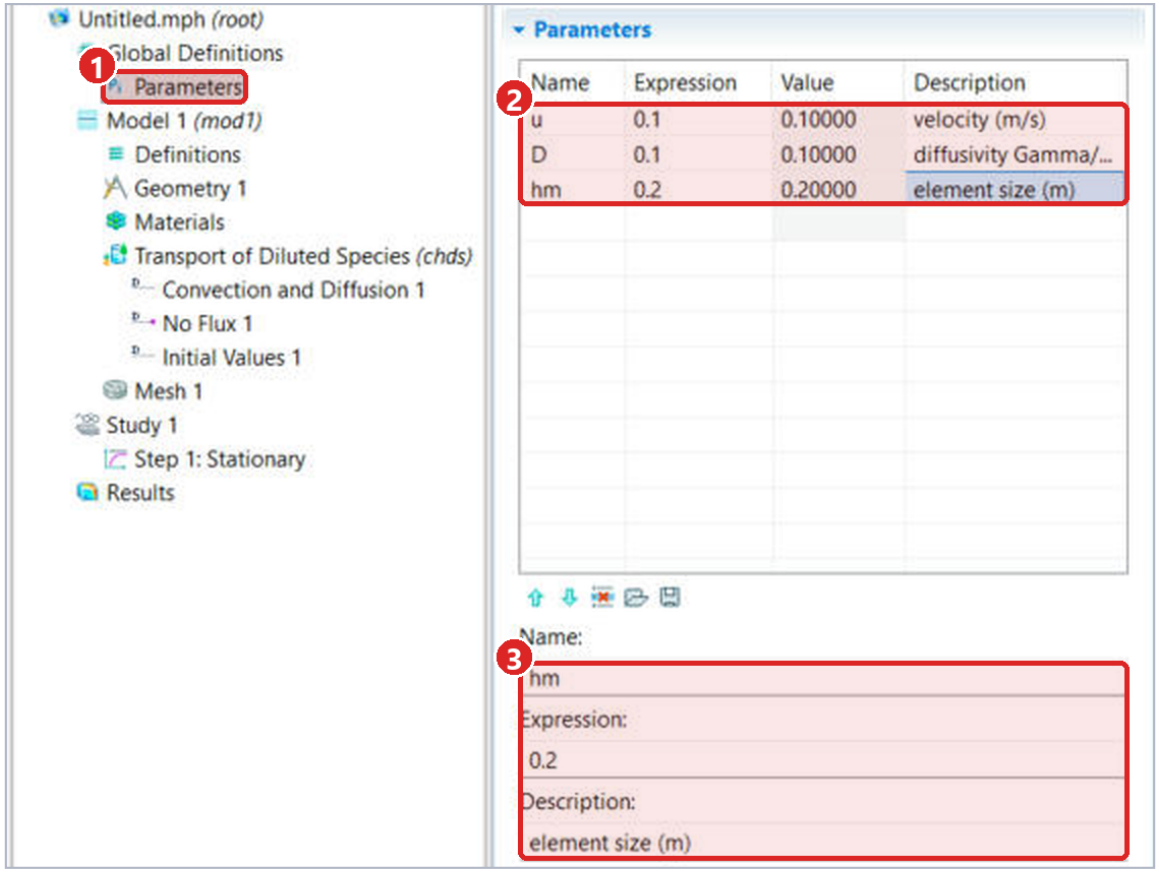

1. Global Definitions: we store u, D and the element size as parameters here.
2. Geometry 1: the line 0 <= x <= 1 m.
3. Transport of Diluted Species (chds): the equation. We will also switch off its numerical stabilisation.
4. Convection and Diffusion 1: we give the velocity u and the diffusion coefficient D.
5. No Flux 1: the default on every boundary. We override both end points with Concentration conditions.
6. Mesh 1: the grid. 5 elements of 0.2 m give the grid of cases 1 and 2.
7. Study 1 with Step 1: Stationary.

**Why?**
There is no material and no source: only u, D and the two end values are needed.

# Step 5 - Enter the problem data as parameters



*Parameters keep the numbers in one place, so the sweep over u and the element size is easy.*

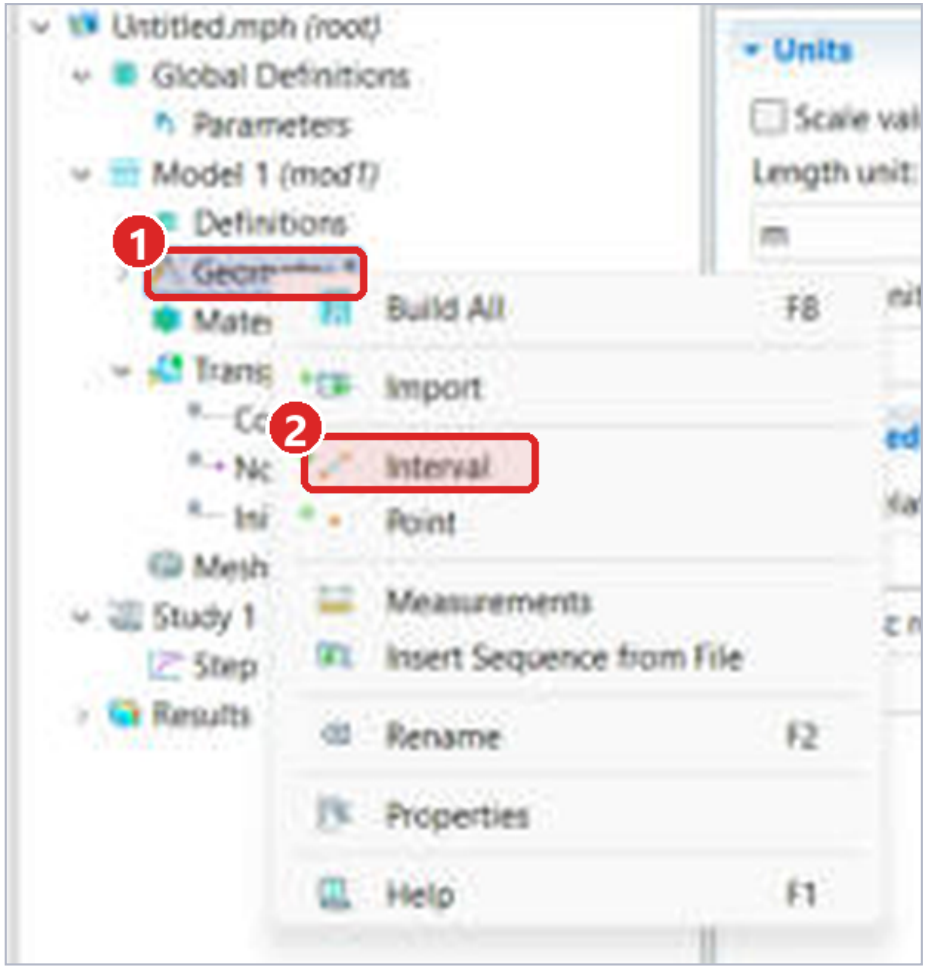


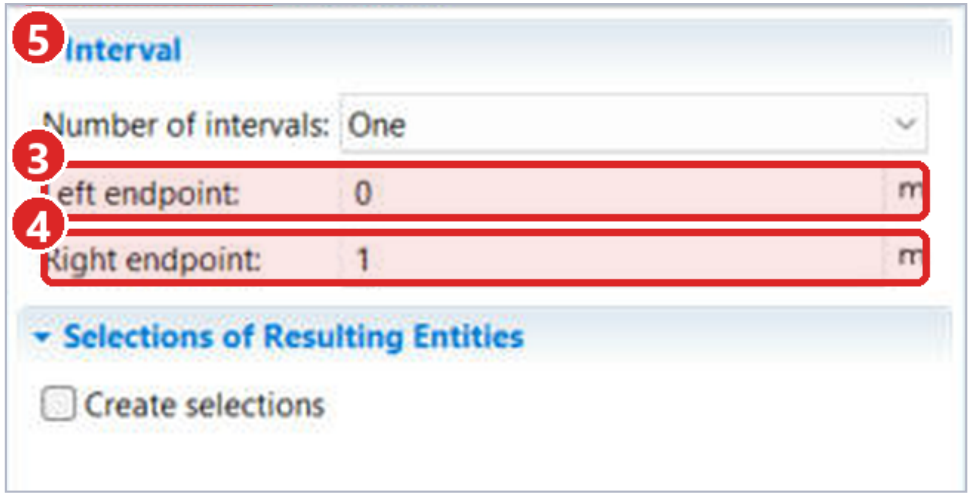

1. Right-click Global Definitions > Parameters (left-click Global Definitions first).
2. Three rows: u = 0.1 (velocity, m/s); D = 0.1 (diffusivity Gamma / rho, m^2/s); hm = 0.2 (element size, m). Type the Name, press Tab, type the Expression, press Tab, type the Description, press Enter.
3. The fields below the table show the selected row (here hm = 0.2).

**Why?**
D = Gamma / rho = 0.1 kg/(m s) / 1 kg/m^3 = 0.1 m^2/s. The cell Peclet number is Pe = u hm / D, so u = 2.5 and hm = 0.2 give Pe = 5. We will sweep u and hm in Step 13.

# Step 6 - Build the geometry: a line from 0 to 1 m

*The domain is the interval 0 <= x <= L = 1 m.*




1. Right-click Geometry 1 (left-click it first if no menu appears).
2. Choose Interval.
3. Left endpoint = 0 (m).
4. Right endpoint = 1 (m): the defaults already give L = 1 m.
5. Click Build Selected (or press F8).

# Step 6 - Build the geometry: a line from 0 to 1 m



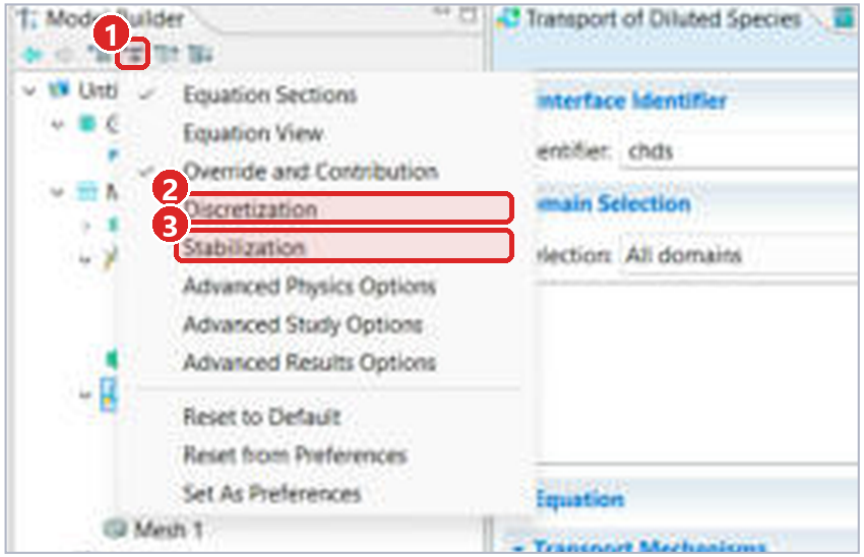

6 The line appears, x from 0 to 1 m. Point 1 is the left end (phi = 1), point 2 the right end (phi = 0).

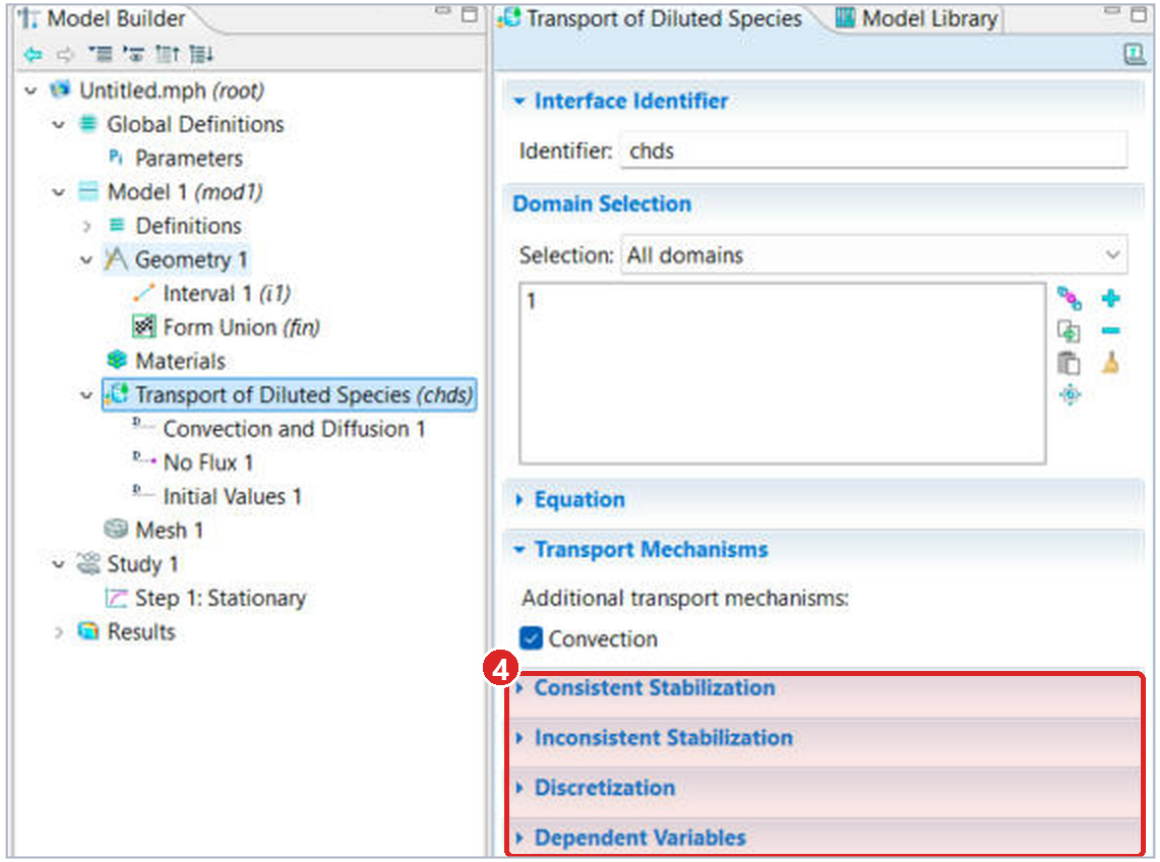
**Why?**
COMSOL numbers the end points 1 (left) and 2 (right). We use these numbers for the boundary conditions.

# Step 7 - Show the stabilisation settings



*Commercial codes add numerical stabilisation to the convection term by default. To compare with the plain central difference scheme of the problem we must switch it off. The settings are hidden at first. First select the node Transport of Diluted Species.*



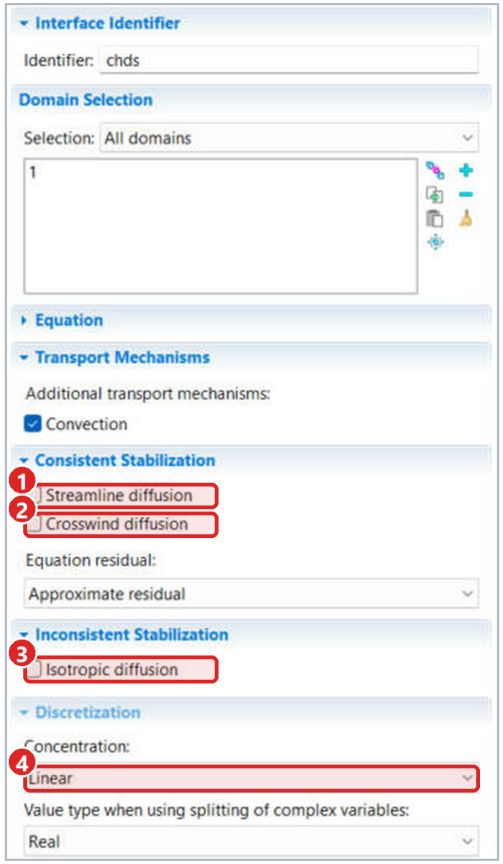




1. Click the Show button (the funnel icon, fourth in the Model Builder toolbar).
2. Tick Discretization.
3. Tick Stabilization.
4. Three new sections appear in the settings window: Consistent Stabilization, Inconsistent Stabilization and Discretization.

**Why?**
Without this step the stabilisation options stay hidden and COMSOL uses its defaults: streamline diffusion and crosswind diffusion switched on.

# Step 8 - Switch the stabilisation off

*With the default stabilisation COMSOL adds artificial diffusion, much like the upwind scheme. We switch it off to get the central difference scheme.*

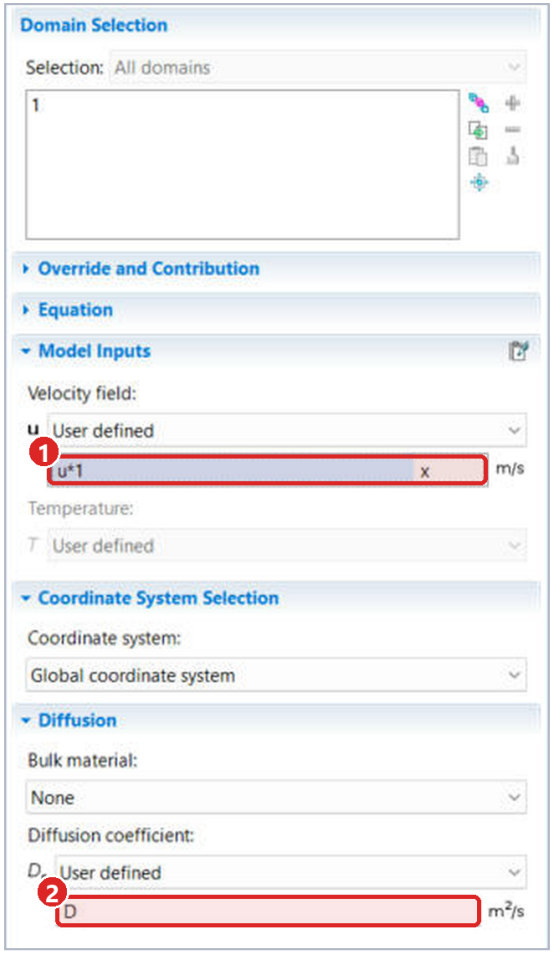



1. Consistent Stabilization: untick Streamline diffusion.
2. Untick Crosswind diffusion.
3. Inconsistent Stabilization: Isotropic diffusion stays unticked (default).
4. Discretization: Concentration = Linear (default). Linear elements on a uniform 1D mesh give exactly the central difference scheme.

**Why?**
With linear elements and a uniform mesh the finite element equations for an interior node are -(D/dx + u/2) c(i-1) + 2 D/dx c(i) - (D/dx - u/2) c(i+1) = 0: identical to the central difference equations of the problem. Step 16 shows what the default stabilisation does.

# Step 9 - Velocity and diffusion coefficient

*We give COMSOL the velocity field and the diffusivity (no flow equations are solved).*

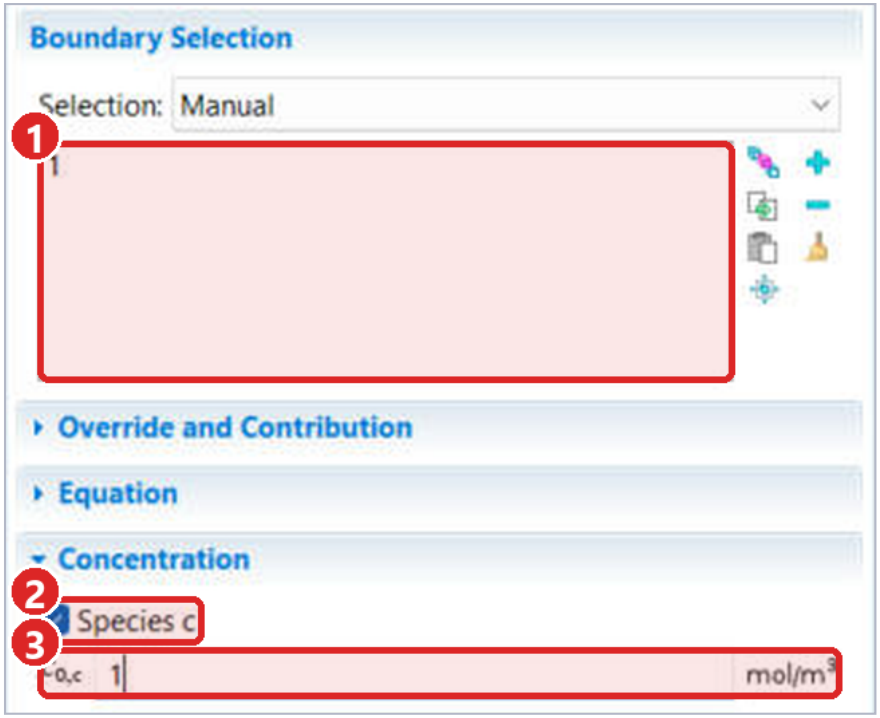


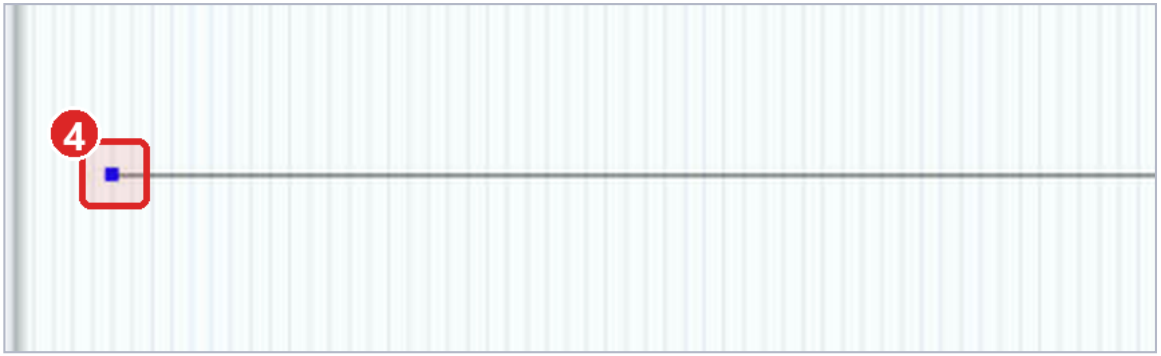

1. Velocity field u: User defined, x-component u (the parameter). Type u*1 to avoid that COMSOL autocompletes u to ua. Unit m/s.
2. Diffusion coefficient D_c: User defined, D (the parameter, m^2/s).

**Why?**
The velocity is constant, so continuity d(rho u)/dx = 0 holds automatically. The convective flux is u c; the diffusive flux is -D dc/dx.

# Step 10 - Left end: concentration 1

*The left reservoir keeps phi = phi_0 = 1.*



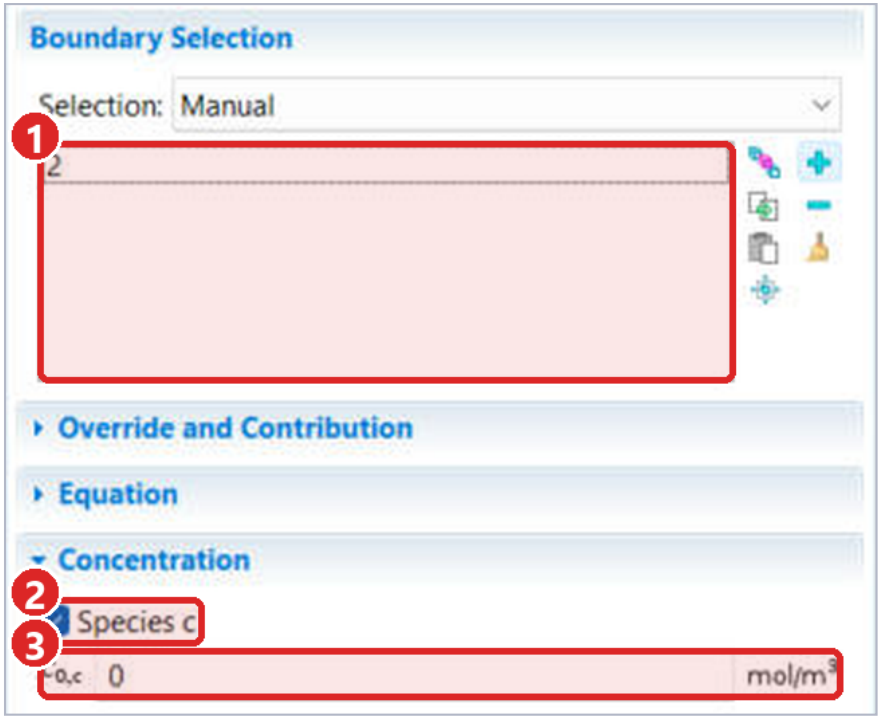

1. Right-click Transport of Diluted Species > Concentration. The selection list must contain 1 (the left end point): hover over the point until it turns red, click it, press +.
2. Tick Species c.
3. Concentration c0,c = 1 mol/m^3.
4. The left end point is drawn in blue.

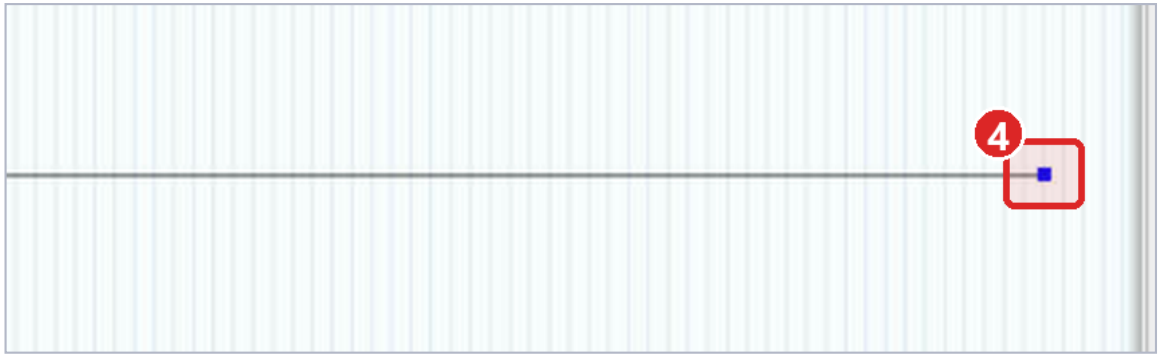
**Why?**
This is the Dirichlet condition phi(0) = 1.

# Step 11 - Right end: concentration 0



*The right reservoir holds phi = phi_L = 0.*



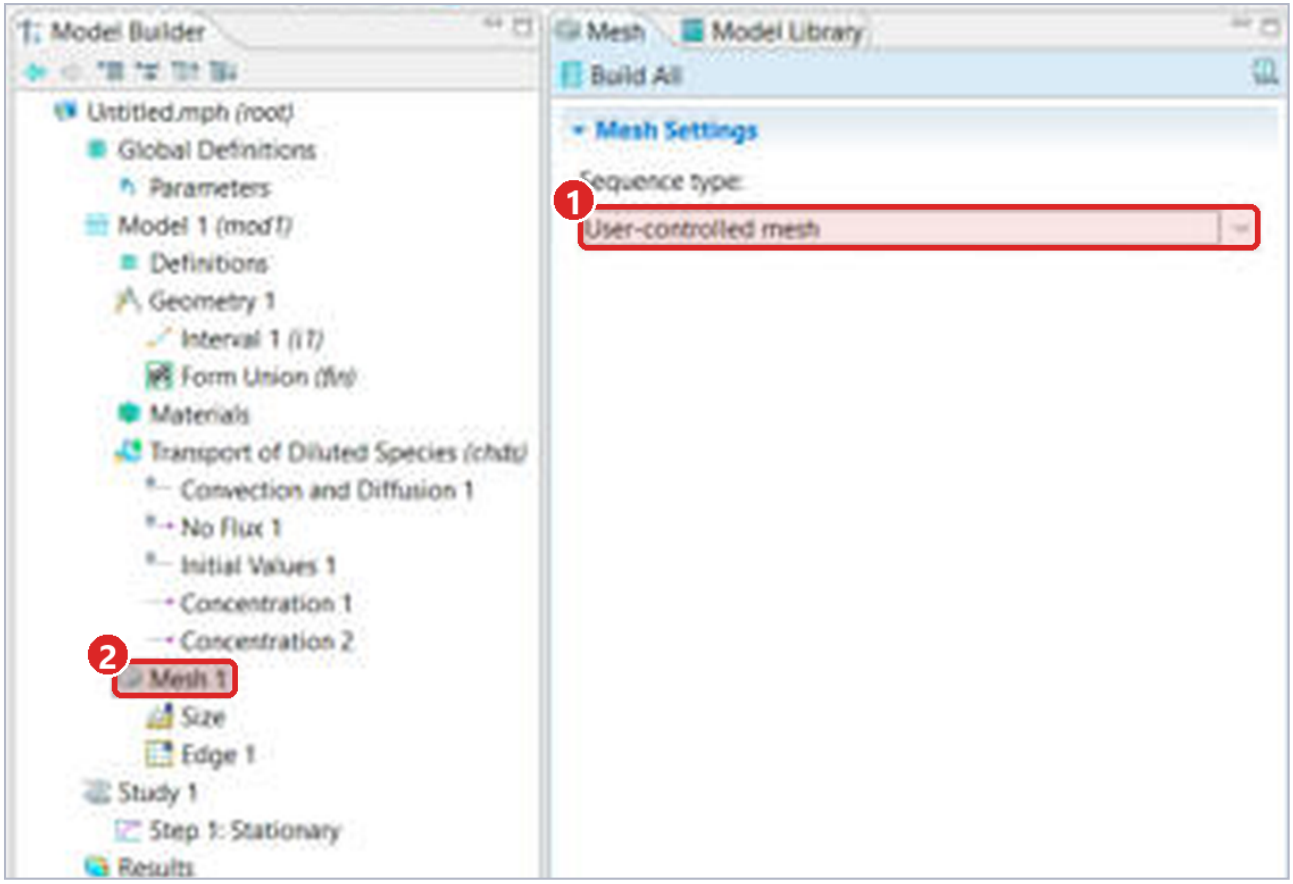

1. Add a second Concentration node. Its selection list must contain 2 (the right end point). If a wrong point is listed, select it in the list and press the minus button.
2. Tick Species c.
3. Concentration c0,c = 0 mol/m^3.
4. The right end point is drawn in blue.

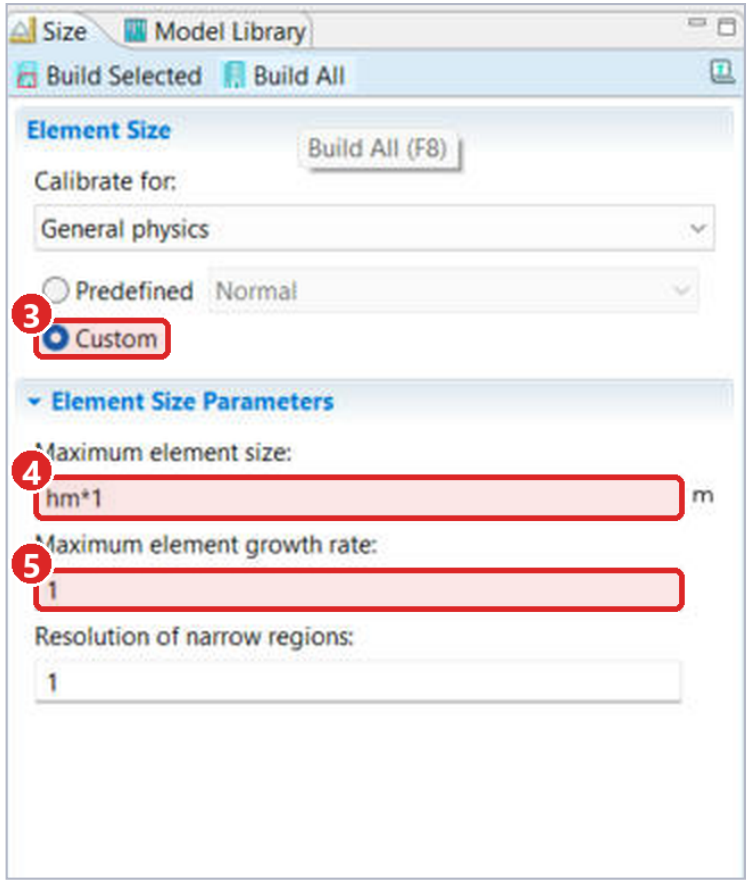
**Why?**
This is the Dirichlet condition phi(L) = 0. Together with Step 9 the model has the two boundary values of the problem.

# Step 12 - Create the mesh with element size hm



*A uniform mesh of elements of size hm is the equivalent of the finite difference grid.*




1. Click Mesh 1, set Sequence type to User-controlled mesh. The nodes Size and Edge 1 appear.
2. Mesh 1 is selected.
3. Click the node Size and select Custom.
4. Maximum element size = hm*1 (the parameter hm).
5. Maximum element growth rate = 1: this makes all elements equal. Then click Build All (F8).

# Step 12 - Create the mesh with element size hm



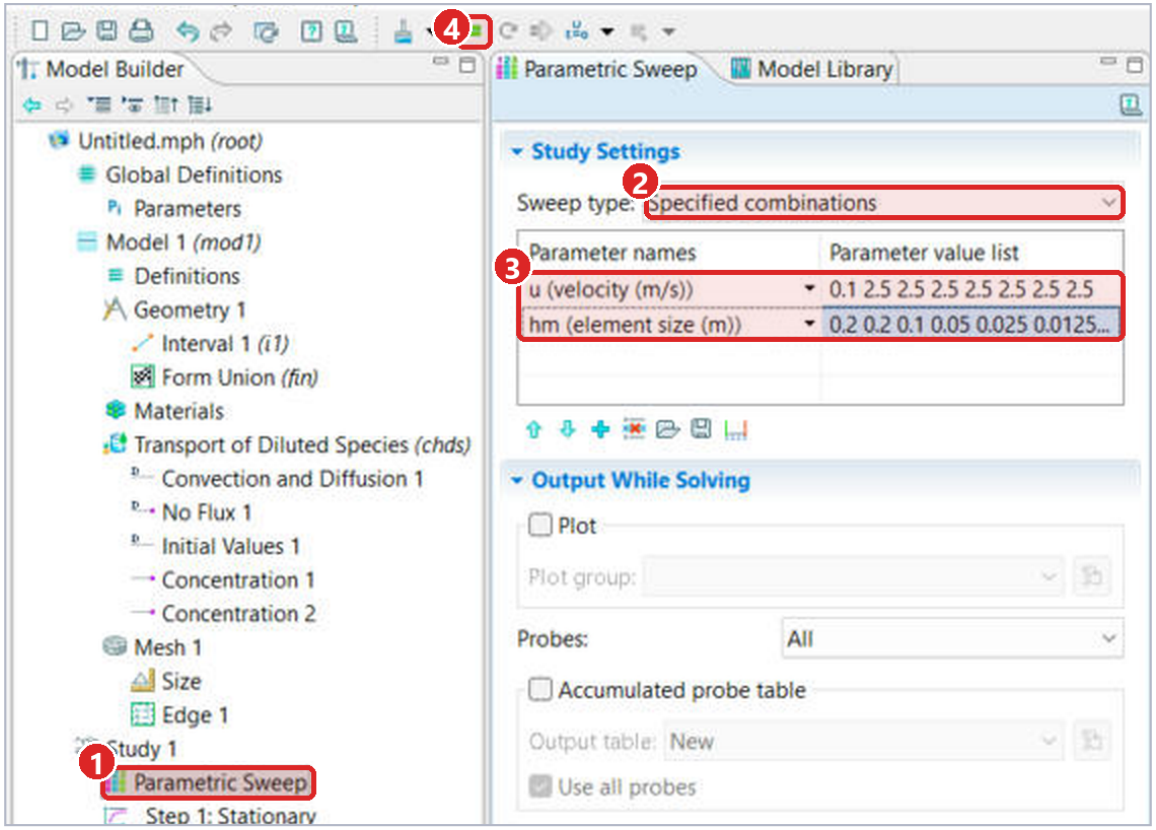

6 Messages: Complete mesh consists of 5 elements (hm = 0.2 m).

**Why?**
With hm = 0.2 the 1 m line is divided into 5 elements of 0.2 m: the grid of cases 1 and 2 (nodes 0 to 5). The sweep will set hm = 0.1, 0.05, ... for case 3.

# Step 13 - Add a parametric sweep over u and hm



*One study run solves all cases: case 1, case 2 and the eight grids of case 3. A parametric sweep with specified combinations pairs the value of u with the value of hm.*

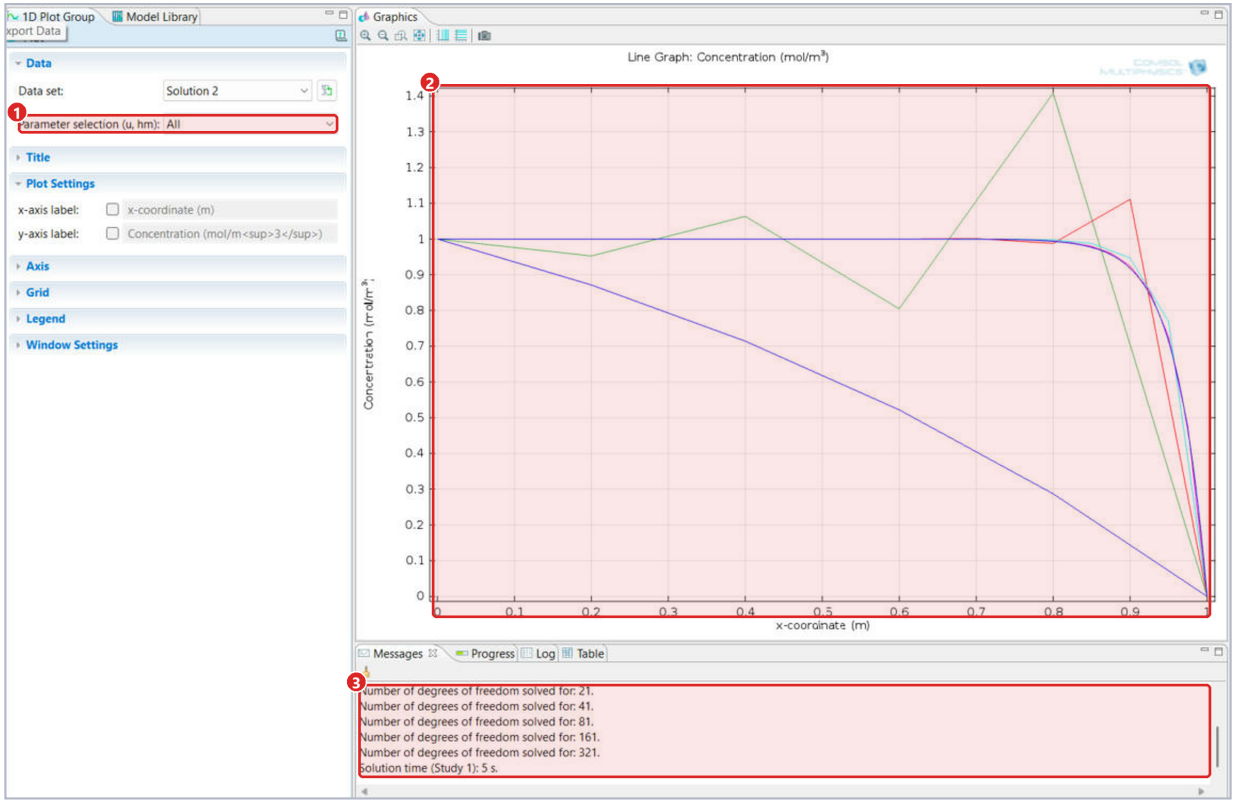



1. Right-click Study 1 > Parametric Sweep (left-click Study 1 first).
2. Sweep type: Specified combinations.
3. Row 1: u with the list 0.1 2.5 2.5 2.5 2.5 2.5 2.5 2.5. Row 2: hm with the list 0.2 0.2 0.1 0.05 0.025 0.0125 0.00625 0.003125. Add the second row with the + button below the table and choose hm in the parameter-name drop-down (it offers D first). The k-th value of u goes with the k-th value of hm.
4. Click Compute (the green = in the toolbar, or press F8).

**Why?**
Combination 1 = case 1 (u = 0.1, 5 elements). Combination 2 = case 2 (u = 2.5, 5 elements). Combinations 2 to 8 = case 3: u = 2.5 with 5, 10, 20, 40, 80, 160 and 320 elements.

# Step 14 - Compute and look at the profiles

*After Compute COMSOL plots the concentration along x for all eight combinations.*



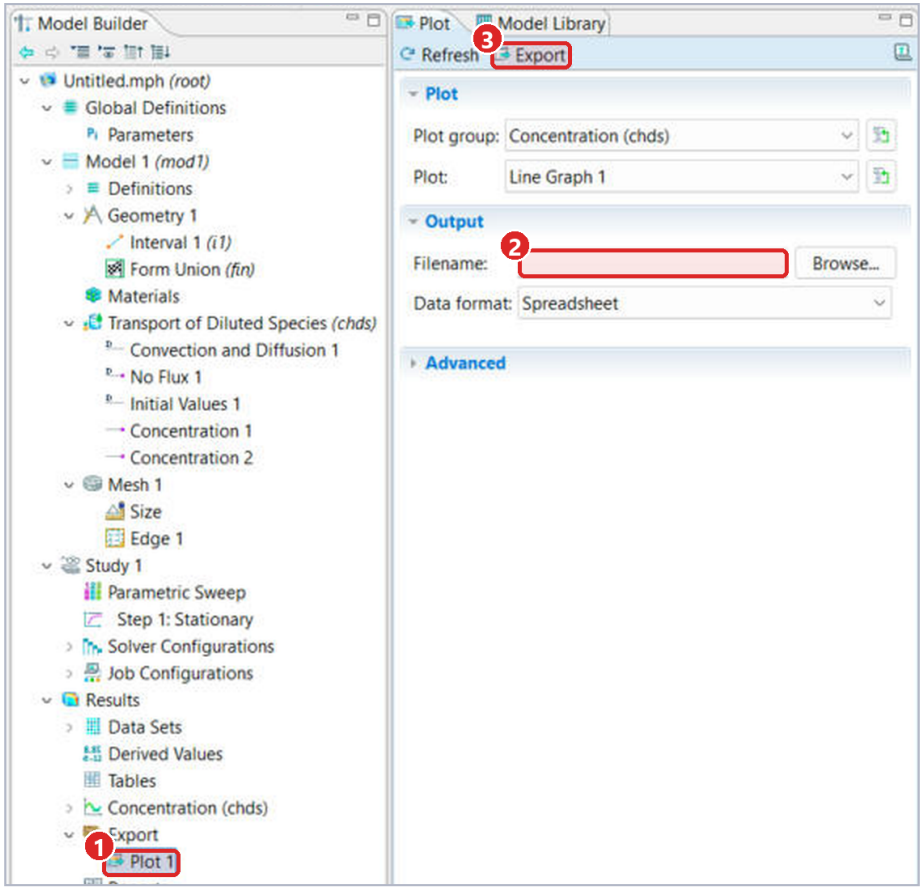

1. Parameter selection (u, hm): All. The plot shows the eight solutions of Data set Solution 2.
2. The straight blue curve decreasing from 1 to 0 is case 1 (u = 0.1). The curve with big wiggles (peak 1.41 at x = 0.8, dip 0.80 at x = 0.6) is case 2: central differences with 5 elements, which is not physical (phi cannot exceed 1). The next curve (peak 1.11 at x = 0.9) has 10 elements. The finer grids are smooth and stay between 0 and 1.
3. Messages: degrees of freedom 6, 6, 11, 21, 41, 81, 161, 321 (one per node), Solution time (Study 1): 5 s.

**Why?**
A concentration above 1 or below 0 is impossible: the exact solution lies between the two reservoir values. The wiggles are a numerical artefact of the central difference scheme when the cell Peclet number is larger than 2 (here 5).

# Step 15 - Export the profiles



*To read the nodal values and compute the errors we save the plotted data in a text file.*

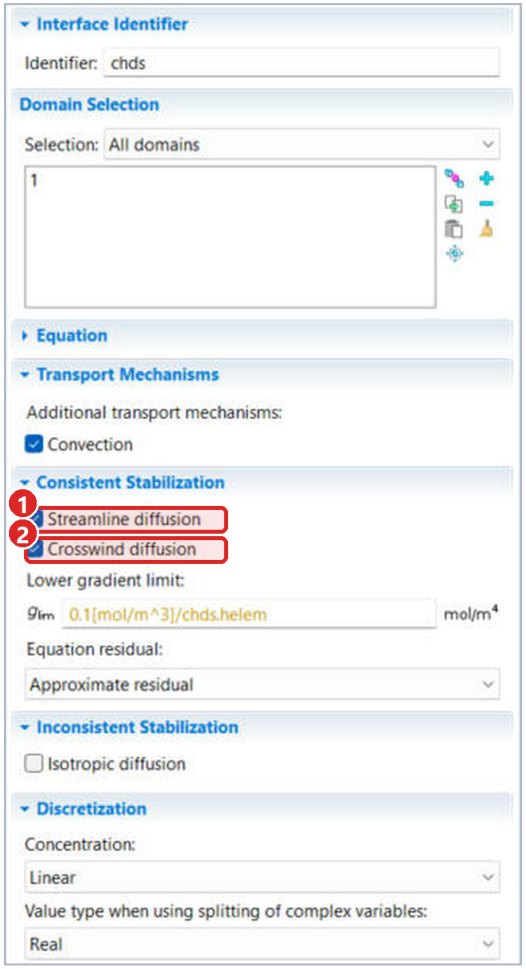



1. Right-click Results > Export > Plot. The node Plot 1 appears (plot group Concentration (chds), Line Graph 1).
2. Filename: type the path of a text file you can write to (here profiles.txt).
3. Click Export. The Messages window confirms 'Exported plot data'.

**Why?**
The file holds x and c for all eight solutions one after the other (each element of the mesh is subdivided for plotting; the mesh nodes are among the points). The accompanying Python script reads the values at the mesh nodes and compares them with the exact solution.

# Step 16 - Compare with COMSOL's default stabilisation

*To see what commercial codes do by default, tick the two stabilisation options again and recompute.*




1. Tick Streamline diffusion (the default).
2. Tick Crosswind diffusion (the default).

# Step 16 - Compare with COMSOL's default stabilisation




3 Select Study 1 and press F8. Now all eight curves stay between 0 and 1: the wiggles are gone, but the steep layer at x = 1 is smeared (the node values are too low near x = 0.8).

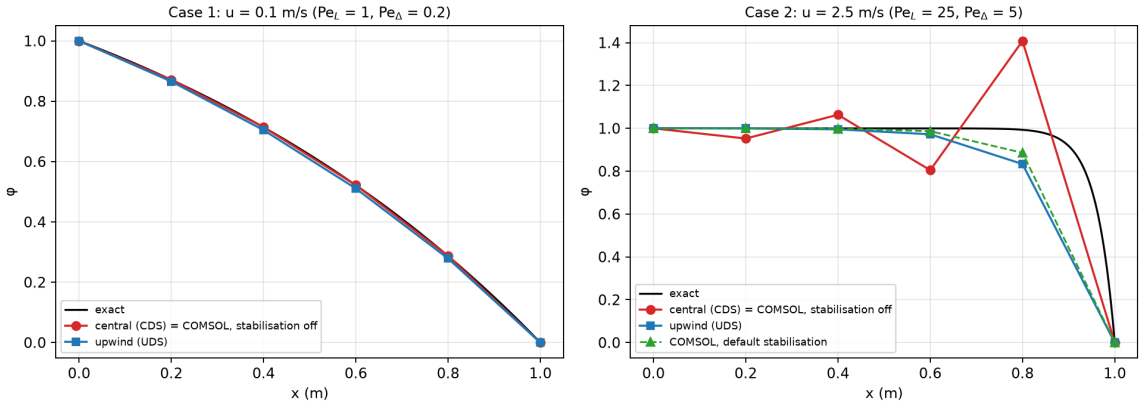
**Why?**
Streamline diffusion adds artificial diffusion in the flow direction, as the upwind scheme does, but only as much as needed. At 5 elements the largest nodal error is 0.107, less than central (0.415) and upwind (0.160); for fine grids all schemes converge to the same profile.

# Results: COMSOL against the finite difference schemes

## A. Nodal values, N = 5 (Dx = 0.2 m)

| **Case 1: u = 0.1 m/s** | **x=0** | **0.2** | **0.4** | **0.6** | **0.8** | **1.0** | **max error** |
|---|---|---|---|---|---|---|---|
| exact | 1.0000 | 0.8711 | 0.7138 | 0.5215 | 0.2868 | 0.0000 | |
| central (CDS) | 1.0000 | 0.8714 | 0.7141 | 0.5220 | 0.2871 | 0.0000 | 0.0004 |
| upwind (UDS) | 1.0000 | 0.8656 | 0.7044 | 0.5109 | 0.2786 | 0.0000 | 0.0107 |
| COMSOL, stabilisation off | 1.0000 | 0.8714 | 0.7141 | 0.5220 | 0.2871 | 0.0000 | 0.0004 |
| **Case 2: u = 2.5 m/s** | **x=0** | **0.2** | **0.4** | **0.6** | **0.8** | **1.0** | **max error** |
| exact | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9933 | 0.0000 | |
| central (CDS) | 1.0000 | 0.9525 | 1.0633 | 0.8047 | 1.4082 | 0.0000 | 0.4149 |
| upwind (UDS) | 1.0000 | 0.9994 | 0.9955 | 0.9723 | 0.8334 | 0.0000 | 0.1598 |
| COMSOL, stabilisation off | 1.0000 | 0.9525 | 1.0633 | 0.8047 | 1.4082 | 0.0000 | 0.4149 |
| COMSOL, default stabilisation | 1.0000 | 0.9999 | 0.9985 | 0.9870 | 0.8860 | 0.0000 | 0.1073 |

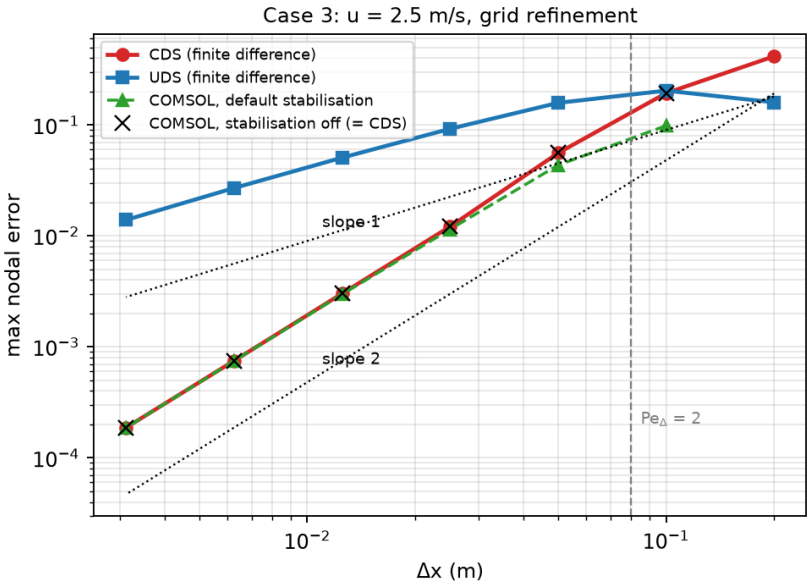
COMSOL with the stabilisation switched off equals the central difference solution to 1e-15 in both cases. At Pe = 0.2 (case 1) all schemes agree with the exact solution within 0.011. At Pe = 5 (case 2) the central scheme gives phi = 1.41 at x = 0.8 (above both reservoir values), CDS has a negative coefficient a_E = -0.75, the upwind scheme is bounded but smears the front, and COMSOL's default stabilisation is bounded and closer to the exact profile.

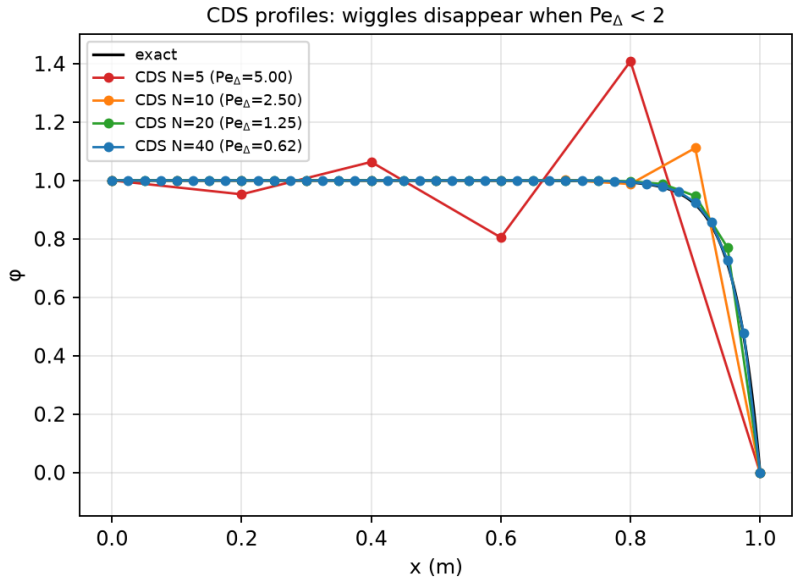


# Grid study: when does the central scheme work, and what is the order?

## Case 3: u = 2.5 m/s, maximum error at the nodes

| N | Dx (m) | Pe_Delta | CDS error | order | UDS error | order | COMSOL off | COMSOL stab. | CDS wiggles |
|---|---|---|---|---|---|---|---|---|---|
| 5 | 0.20000 | 5.00 | 4.15e-01 | | 1.60e-01 | | 4.15e-01 | 1.07e-01 | yes |
| 10 | 0.10000 | 2.50 | 1.93e-01 | 1.10 | 2.04e-01 | -0.35 | 1.93e-01 | 9.83e-02 | yes |
| 20 | 0.05000 | 1.25 | 5.57e-02 | 1.79 | 1.58e-01 | 0.37 | 5.57e-02 | 4.34e-02 | no |
| 40 | 0.02500 | 0.62 | 1.21e-02 | 2.20 | 9.22e-02 | 0.78 | 1.21e-02 | 1.15e-02 | no |
| 80 | 0.01250 | 0.31 | 3.02e-03 | 2.01 | 5.07e-02 | 0.86 | 3.02e-03 | 2.97e-03 | no |
| 160 | 0.00625 | 0.16 | 7.49e-04 | 2.01 | 2.70e-02 | 0.91 | 7.49e-04 | 7.46e-04 | no |
| 320 | 0.00313 | 0.08 | 1.87e-04 | 2.00 | 1.39e-02 | 0.95 | 1.87e-04 | 1.87e-04 | no |




COMSOL's columns start at N = 5: stabilisation off = central scheme; 'stab.' = default streamline and crosswind diffusion. The error of the central scheme falls by a factor 4 per halving of Dx (order 2) once the grid is fine enough; the upwind error only by a factor 1.9 (order 1).

# Answers to the tasks

**Task 1, physics.**

Convection carries phi downstream with the flow; diffusion spreads it from high to low values. Pe_L = rho u L / Gamma = 1 (u = 0.1) and 25 (u = 2.5). For large u the exact solution is flat (phi close to 1) with a thin boundary layer at x = L.

**Task 2, grid.**

N = 5 intervals, Dx = 0.2 m, nodes 0 to 5, unknowns 1 to 4. Cell Peclet number Pe_Delta = rho u Dx / Gamma = 0.2 and 5.

**Task 3, discretisation.**

CDS: a_W = D + F/2, a_E = D - F/2, a_P = 2D. UDS: a_W = D + F, a_E = D, a_P = 2D + F, with F = rho u and D = Gamma/Dx.

**Task 4, solution.**

Thomas algorithm (forward elimination, back substitution) for the tridiagonal system, worked by hand on page 4.

**Task 5, physically possible?**

Case 1: both schemes are good (CDS error 4e-4, UDS 0.011). Case 2: CDS gives wiggles with phi = 1.41 > 1, which is impossible for a quantity transported without sources; its coefficient a_E = -0.75 is negative. UDS is always bounded because all its coefficients are positive, but it smears the front.

**Task 6, false diffusion.**

UDS = CDS with the extra diffusion rho u Dx / 2: 0.01 (10 % of Gamma) in case 1 and 0.25 (250 % of Gamma) in case 2. This numerical diffusion is the price for boundedness and gives a first-order scheme.

**Task 7, grid study.**

The central scheme starts to work when Pe_Delta < 2, i.e. Dx < 0.08 m: N = 13 is the first grid without wiggles (N = 12 still has them); among the grids of the study it is N = 20. Orders: CDS second order (measured 2.0), UDS first order (0.95 at N = 320). On coarse grids the upwind scheme can even be more accurate (N = 5: 0.16 against 0.41).

**Task 8, Gauss-Seidel.**

UDS converges (spectral radius 0.32). CDS diverges (3.44) because a_E < 0 destroys diagonal dominance.

**COMSOL.**

With stabilisation switched off, linear elements on a uniform mesh reproduce the central scheme exactly. COMSOL's default stabilisation (streamline + crosswind diffusion) keeps the solution bounded, at 5 elements with a smaller error (0.107) than both UDS and CDS.

# Troubleshooting and tips

**The right-click menu does not appear.**

Left-click the node once to select it, then right-click again.

**A selection list stays empty or contains the wrong point.**

Hover the cursor over the end point until it turns red, click it, then press the + button. If the list shows a point you did not want, select it in the list and press the minus button. Points: 1 = left end, 2 = right end.

**Typing in a field added to the old text.**

Click in the field, press Ctrl+A to select everything, then type the new value.

**The velocity field shows ua instead of u.**

COMSOL autocompletes the name. Type u*1 (or choose u from the parameter list) so that the expression is exactly the parameter.

**I cannot find the stabilisation settings.**

Select the node Transport of Diluted Species, click the Show button in the Model Builder toolbar and tick Stabilization and Discretization.

**My COMSOL solution has no wiggles at 5 elements.**

The default stabilisation is still switched on. Untick Streamline diffusion and Crosswind diffusion (Step 7).

**The second row of the sweep table shows D instead of hm.**

Open the drop-down in the Parameter names column of that row and choose hm.

**The mesh has a different number of elements than expected.**

Set the Maximum element growth rate to 1 so that all elements have the size hm.

**COMSOL asks for a license.**

Start COMSOL with the shortcut your installation provides. On campus the computer must be on the university network or VPN.

**Save your work.**

File > Save As, choose a folder you can write to. Compare with the supplied file convdiff_comsol.mph.